

Programme	B. Sc. Mathematics Honours			
Course Code	MAT5EJ302(1)			
Course Title	DATA STRUCTURES AND ALGORITHMS			
Type of Course	Elective (Specialisation- Mathematical Computing)			
Semester	V			
Academic Level	300 - 399			
Course Details	Credit	Lecture/Tutorial per week	Practical per week	Total Hours
	4	4	-	60
Pre-requisites	1. Fundamental Mathematics Concepts: Sets, Functions 2. Discrete Mathematics			
Course Summary	This course familiarises students with computational problems and computational thinking using some of the basic algorithmic strategies.			

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Analyse and compare the efficiency of algorithms for computing Fibonacci numbers, distinguishing between exponential and polynomial approaches.	E	P	Internal Exam/Assignment/ Seminar/ Viva / End Sem Exam
CO2	Demonstrate proficiency in asymptotic analysis to assess the efficiency of algorithms.	Ap	P	Internal Exam/Assignment/ Seminar/ Viva / End Sem Exam
CO3	Apply classical algorithms for number operations, including addition, multiplication, and modular arithmetic, to solve computational problems efficiently.	Ap	P	Internal Exam/Assignment/ Seminar/ Viva / End Sem Exam
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Text Book		<i>Algorithms</i> by Sanjoy Dasgupta, Christos H. Papadimitriou, Umesh Vazirani. McGraw- Hill Education, 2006. ISBN: 978-0073523408.		
Module	Unit	Content	Hrs (48+12)	Ext. Marks (70)
I	Introduction		12	
	1	Computing Fibonacci Numbers: Exponential and Polynomial Algorithms		
	2	Efficiency of Algorithms: Asymptotic Analysis, Big-O Notation		
	3	Algorithms with Numbers: Efficiency of classical Addition and Multiplication algorithms		
	4	Algorithms for Modular Arithmetic		
	5	Euclid’s Algorithm for GCD		
	6	Primality Testing		
	<i>Sections from Text: 0.2, 0.3, 1.1, 1.2, 1.3</i>			
II	Divide and Conquer Algorithms and Graph Search		12	
	7	Fast Integer Multiplication		
	8	Recursive Relations		
	9	Binary Search		
	10	Merge Sort		
	11	Graph Representations: Adjacency Matrix, Adjacency List		
	12	Depth First Search Undirected Graphs		
	13	Depth First Search in Directed Graphs		
	<i>Sections from Text: 2.1, 2.2. 2.3, 3.1-3.3.</i>			
III	Graph Algorithms		12	
	14	Checking connectivity		
	15	Directed Acyclic Graphs, Strongly Connected Components		
	16	Breadth First Search and Computation of distances.		
	17	Weighted Graphs and Dijkstra’s Algorithm		
	18	Priority queue implementations		
	19	Shortest Paths in Directed Acyclic Graphs		

	<i>Sections from Text: 3.4, 4.1 to 4.4, 4.5, 4.7</i>		
IV	Greedy & Dynamic Programming Algorithms		12
	20	Minimum Spanning Trees: Cut Property	
	21	Kruskal's Algorithm	
	22	Data structure for disjoint sets.	
	23	Prim's algorithm	
	24	Dynamic Programming and Shortest Path in Directed Acyclic Graphs (DAG)	
	25	All pairs of Shortest Paths and Floyd Warshall Algorithm	
	<i>Sections from Text: 5.1, 5.4, 6.1, 6.6.</i>		
V (Open Ended)	Advanced Topics		12
	27	Implement the following algorithms in Python - Fibonacci Numbers (exponential and polynomial) - Euclid's algorithm (extended version) - Primality Testing - Depth First Search (and checking connectivity) - Breadth First Search (and calculating distances) - Dijkstra's Algorithm	
References: 1. <i>The Design and Analysis of Algorithms</i> by Dexter C Kozen. Texts and Monographs in Computer Science, Springer, 1992. ISBN:0-387-97687-6. 2. <i>Introduction to Algorithms</i> (3rd Edition) by Thomas H. Cormen, Charles E. Leiserson, Ronald L Rivest, Clifford Stein. PHI Learning, 2009. ISBN:978-81-203-4007-7. 3. <i>Algorithm Design</i> by Jon Kleinberg and Eva Tardos. Pearson, 2015. ISBN:978-93-325-1864.			

Note: 70 external marks are distributed over the first four modules subjected to a minimum of 15 marks from each module.

Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO4	PSO5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	2	2	3	2			3	1	3	3	3	0	3
CO 2	2	3	2	2			3	1	3	3	3	0	2
CO 3	2	3	3	2			3	1	3	3	3	0	2

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Assignment/ Seminar
- Internal Exam
- Viva
- Final Exam (70%)

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment	Seminar	Viva	End Semester Examinations
CO 1	✓	✓	✓	✓	✓
CO 2	✓	✓	✓	✓	✓
CO 3	✓	✓	✓	✓	✓